Raspberry Pi Onramp

A Beginner’s Guide to Tinkering with Hardware and Code

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Outline

1. General Overview

Why Use Raspberry Pi

Raspberry Pi is a popular choice for beginners because it is affordable, well-documented, and has a large community. It ships with a familiar Debian-based environment, making it easy to get started with Linux. It is equally suited for advanced users who want to build custom hardware projects, run home servers, or experiment with electronics via its 40-pin GPIO header.

What is Raspberry Pi

Raspberry Pi is a **low-cost** operating systems. It is a popular platform for single-board computer *capable of running* Linux-based embedded systems, IoT projects, home automation, education, and prototyping. It was originally developed by the Raspberry Pi Foundation (now Raspberry Pi Ltd.) with the goal of promoting computer science education.

The official operating system is **Raspberry Pi OS** (formerly called Raspbian until May 2020). It is based on the Debian Linux distribution and uses the `apt` (Advanced Package Tool) package manager.

Before installing any software, it is good practice to refresh the local package index and then apply all available upgrades:

```bash
sudo apt update  # refresh the package list
sudo apt full-upgrade  # upgrade packages, handling dependency changes
```

`full-upgrade` is preferred over `upgrade` because it correctly handles cases where packages need to be added or removed to satisfy updated dependencies.

To install a specific package:

```bash
sudo apt install git  # install git
```

SSH

Secure Shell Configuration

1.1 SSH Setup

SSH (Secure Shell) is a cryptographic network protocol for secure remote access to a machine over an unsecured network. On Raspberry Pi OS, the `openssh-server` package is already installed but **disabled by default** for security reasons.

1.1 SSH Setup

There are three common ways to enable it:

Option 1 — raspi-config (recommended for interactive use):

```bash
sudo raspi-config
# Navigate to: Interface Options → SSH → Enable
```

Option 2 — systemctl (enable on a running system):

```bash
sudo systemctl enable ssh  # enable SSH to start on boot
sudo systemctl start ssh  # start the SSH service immediately
```

Option 3 — headless setup (before first boot):

Create an empty file named `ssh` (no extension) in the `/boot` or `/bootfs` partition of the SD card. The OS will detect it on first boot, enable the SSH service, and delete the file.

1.1 SSH Setup

Once SSH is enabled, connect from another machine with:

```bash
ssh <username>@<hostname>.local
# e.g. ssh pi@raspberrypi.local
```

INFO

Create a new connection profile of type Ethernet on the physical interface enp4s0. The PC will act as a DHCP server and automatically assign an IP address to any device connected via Ethernet (the Pi), while also enabling NAT to share the laptop's internet connection.

```bash
nmcli connection add type ethernet ifname enp4s0 con-name "rpi-share" ipv4.method shared
```

Activate and inspect the connection

```bash
nmcli connection up "rpi-share"
nmcli connection show "rpi-share"
```

Perform a ping sweep across all 254 addresses, without port scanning

```bash
nmap -sn 10.42.0.0/24
```

1.1.1 SSH Key Authentication

Connecting with a password is convenient but less secure. The recommended approach is **public-key authentication**: you generate a key pair on your client machine, then place the public key on the Raspberry Pi. From that point on, no password is needed and brute-force attacks become ineffective.

1.1.2 How it works

A key pair consists of two files:

Private key (`~/.ssh/rpi`) stays on your machine, never shared.

Public key (`~/.ssh/rpi.pub`) copied to the Pi. It can only be used to verify someone who holds the matching private key.

1.1.3 Generate a key pair (on your local machine)

Ed25519 is the current recommended algorithm: it is compact, fast, and more secure than RSA.

```bash
ssh-keygen -t ed25519 -C "your_email@example.com"
```

You will be prompted for:

File location press Enter to accept the default (`~/.ssh/rpi`).

Passphrase strongly recommended. It encrypts the private key on disk so that even if the file is stolen, it cannot be used without the passphrase.

The command produces two files:

```
~/.ssh/rpi      # private key — keep this secret
~/.ssh/rpi.pub  # public key  — this goes on the Pi
```

1.1.4 Copy the public key to the Raspberry Pi

The simplest method is `ssh-copy-id`, which handles directory creation and permissions automatically:

```bash
ssh-copy-id <username>@<pi-ip-address>
# e.g. ssh-copy-id pi@10.42.0.39
```

You will be asked for your Pi password one final time. After this step, the public key is appended to `~/.ssh/authorized_keys` on the Pi.

1.1.5 Connect using the key

```bash
ssh <username>@<pi-ip-address>
# e.g. ssh pi@192.168.1.123
```

SSH automatically finds and uses `~/.ssh/id_ed25519`. If you set a passphrase, you will be prompted for it once. To avoid re-entering it every session, add the key to the SSH agent:

```bash
eval "$(ssh-agent -s)"   # start the agent
ssh-add ~/.ssh/rpi  # load the key (prompts for passphrase once)
```

1.1.6 Disable password login on the Pi

Once key authentication is confirmed to be working, you can disable password logins entirely to harden the Pi against brute-force attacks. Edit the SSH server config:

```bash
sudo vim /etc/ssh/sshd_config
```

Find and set the following lines *(restart the ssh service afterwards)*:

```
PasswordAuthentication no
PubkeyAuthentication yes
```

Then restart the SSH service:

```bash
sudo systemctl restart ssh
```

WARNING

⚠ Do not disable password login until you have verified that key authentication works in a separate terminal session. Locking yourself out would require physical access to the Pi to recover.

Git and GitHub

Version Control and Collaboration

1.2 Git and GitHub

Git is a distributed version control system that tracks changes to files and directories over time. It allows you to commit snapshots of your work, branch off for experiments, and merge changes back together.

GitHub is a cloud-based hosting platform for Git repositories. It adds collaboration features such as pull requests, issues, and Actions on top of standard Git.

To install Git:

```bash
sudo apt install git
```

Basic initial configuration:

```bash
git config --global user.name  "Your Name"
git config --global user.email "you@example.com"
```

To clone an existing repository from GitHub:

```bash
git clone https://github.com/a-mhamdi/rpi-onramp.git
```

Using a key with GitHub

The same key pair `rpi` and `rpi.pub` can authenticate you to GitHub, removing the need for HTTPS tokens when pushing code.

Display your public key and copy the output. Then go to `GitHub` → `Settings` → `SSH and GPG keys` → `New SSH key`, paste it in, and save. Test the connection:

```bash
ssh -T git@github.com
# Expected: Hi <username>! You've successfully authenticated ...
```

To switch an existing local repository from HTTPS to SSH:

```bash
git remote set-url origin git@github.com:<user>/<repo>.git
```

Lazygit

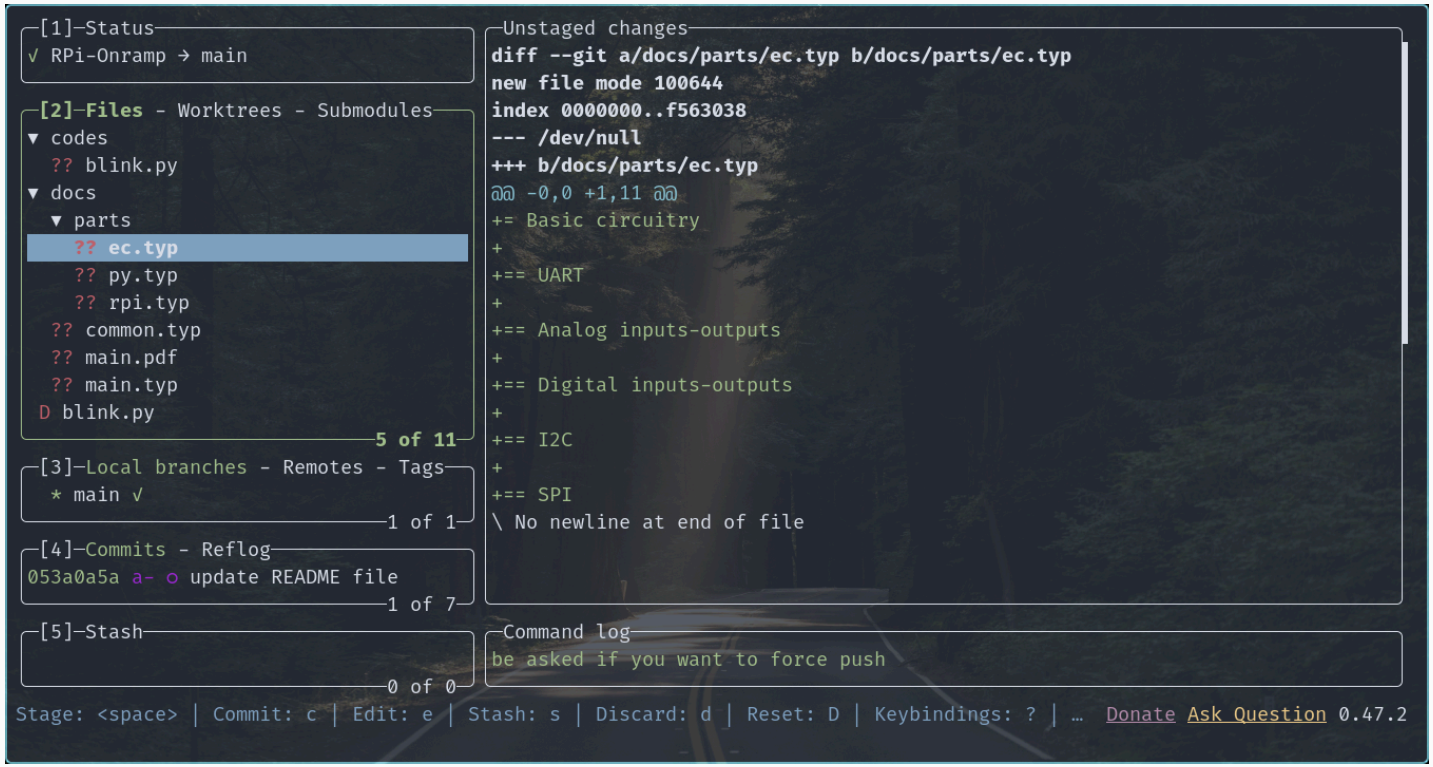
Lazygit is a terminal UI for Git that makes complex operations simple — stage individual lines, perform interactive rebases, and cherry-pick commits, all without memorising obscure flags. It runs directly in your terminal, so there is no need to switch between your editor and a separate GUI application. It is especially useful on the Raspberry Pi where you are often working entirely over SSH with no desktop environment.

To install it on Raspberry Pi OS:

```bash
sudo apt install lazygit
```

Launch it from inside any Git repository by simply running `lazygit`. Press `?` at any time to view the full list of keybindings.

1.2 Git and GitHub



For full documentation and source code, see github.com/jesseduffield/lazygit.

2. Julia Codes

2. Julia Codes

Julia is a high-level, high-performance programming language designed for numerical and scientific computing. While it is often associated with workstations and servers, Julia runs well on ARM-based hardware including the **Raspberry Pi**. This makes it an attractive choice for edge computing, data logging, sensor fusion, and lightweight network services running directly on embedded Linux systems.

This document covers how to install Julia on a **Raspberry Pi**, how to manage GPIO and serial connections, how to use Julia's networking stack, and how to write practical programs that tie these capabilities together.

Getting Started with Julia on Raspberry Pi

Installation and First Steps

2.1.1 Hardware Requirements

Julia can run on any Raspberry Pi model that supports a 64-bit ARM operating system, starting from the Raspberry Pi 3. The Raspberry Pi 4 or 5 is recommended for interactive use and package compilation due to its faster CPU and larger RAM. A microSD card of at least 16 GB is advisable, as Julia's package depot and precompiled artifacts can occupy several gigabytes.

2.1.2 Operating System

Install the 64-bit version of Raspberry Pi OS (formerly Raspbian). Julia's official ARM builds target `aarch64`, so the 64-bit OS is required to use them directly. You can download the image from the Raspberry Pi website and flash it with Raspberry Pi Imager.

After first boot, run a full system update before installing Julia:

```bash
sudo apt update && sudo apt upgrade -y
```

2.1.3 Installing Julia

The recommended way to install Julia on Raspberry Pi is through `juliaup`, the official Julia version manager.

```bash
curl -fsSL https://install.julialang.org | sh
```

Follow the prompts. After installation, open a new shell session or source your profile:

```bash
source ~/.bashrc
```

Verify the installation:

```bash
julia --version
```

`juliaup` allows you to maintain multiple Julia versions and switch between them with

```Bash
juliaup default <version>
```

INFO

If you prefer a simpler path without `juliaup`, the Raspberry Pi OS repositories include a Julia package, though it may be an older version:

```Bash
sudo apt install julia
```

Using the Julia REPL

Interactive Programming and Package Management

2.2 First Launch and the REPL

Start Julia by typing `julia` in the terminal. You will be greeted by the interactive REPL *(Read-Eval-Print Loop)*. From here you can enter expressions, manage packages, and run scripts.

Julia has four REPL modes:

Julia mode *(default)* execute Julia expressions.
Package mode *(press])* manage packages with `add`, `rm`, `status`, `update`.
Help mode *(press ?)* display documentation for any symbol.
Shell mode *(press ;)* run shell commands without leaving Julia.

2.2.1 Package Depot and Precompilation

To install a package, enter package mode and type:

```
1  ] add Plots
```
Julia

To exit the REPL:

```
1  exit()
```
Julia

Julia compiles packages on first use. On a Raspberry Pi this can be slow — several minutes for large packages — but subsequent launches are fast. Precompilation happens automatically when you `add` a package or `using` it for the first time.

2.2.2 Julia Onramp

</> Source code is available at [codes/julia-onramp.ipynb]

Thank you for your attention

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